



Data Integrity Audit & Optimization

Dataset: Dirty Data For Cleaning

Source: Kaggle

Author: Krithika

Link: <https://www.kaggle.com/datasets/krithi96/dirty-data-for-cleaning>

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Introduction:

Raw data is often messy, but businesses need it to make good decisions. To turn a 'dirty' dataset into a useful tool, you need a step-by-step plan that fixes errors without losing important information.

In this project, I used the same scientific mindset I developed during my Physics degree to clean **22,000** hotel booking records. I focused on Data Integrity, making sure every date, price, and city was checked for accuracy and logic. The result is a clean, reliable dataset that is ready for analysis and professional reporting.

Technical Stack:

Tools: Microsoft Excel (Logic & Data Auditing)

Key Techniques: Nested Logical Functions, Statistical Frequency Distribution (for Review Scores), and Attribute-Level Nulling.

Objectives:

- Data Acquisition:** Source a large-scale "dirty" dataset containing multi-source inconsistencies to simulate real-world data challenges.
- Systematic Audit & Cleaning:** Execute a methodical cleaning workflow in Microsoft Excel, utilizing advanced logical functions to resolve errors in dates, pricing, and guest information.
- Optimization for Analysis:** Transform raw, unreliable records into a high-fidelity, standardized dataset ready for professional reporting and dashboarding.

Initial Inspection (Understanding the Data)

Before applying any cleaning methods, I conducted an initial audit to understand the dataset's structure and quality. The data consists of **22,000** hotel booking records across various U.S. cities.

Dataset Summary

Volume: 22,000 Rows | 13 Columns

Scope: Guest identity, booking logistics (dates/nights), financial metrics (price), and operational status (payment/cancellation).

Raw Data

Dataset: Dirty Data For Cleaning

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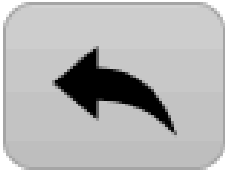
License: CC0: Public Domain

Note : To preserve data integrity, this worksheet is protected. No password is required to unprotect.

booking_id	guest_name	email	listing_id	listing_city	checkin_date	checkout_date	nights	price_per_night	total_price	payment_status	review_score	is_cancelled
BKG-013035		nhoffman@scott.com	1083	Los Angeles	24/05/2023	12/12/2023	-1	245.61	-245.61	pending	6	FALSE
BKG-003115	CAROLYN SHAH	achan@olson-nelson.com	1079	los angeles	19/08/2024	08/11/2024	81	51.96	4208.76		6	Yes
BKG-008732	Jermaine Huang	thomasadam@yahoo.com	1036	San Fran	20/03/2024	03/03/2025	348	383.63	133503.24	paid	1	FALSE
BKG-007591	amy rios	reginald31@robertson-smith.com	1025	LA	29/02/2024	16/03/2025	381	268.36	102245.16	pending	6	n
BKG-000221			1089	LA	26/01/2025	22/02/2025	27	134.71	3637.17	PENDING	N/A	TRUE
BKG-017295		nichole43@gmail.com	1038	LA	03/02/2025	16/02/2025	13	383.22	4981.86	Unpaid	N/A	n
BKG-005857		brianbrown@lynn-taylor.com	1066	San Fran	25/06/2024	19/10/2024	116	283.24	32855.84	N/A	4	TRUE
BKG-012608	Mrs. Amanda Martinez	susanpham@livingston.com	1035	Los Angeles	21/03/2025	28/03/2025	7	189.12	1323.84	PENDING		y
BKG-003463		eharrison@mullins.com	1038	new york	10/08/2023	21/11/2023	103	392.45	40422.35	Unpaid	-1	TRUE
BKG-006478	shelby mason	grahamamanda@sparks.com	1089	San Fran	19/10/2024	14/04/2025	177	195.82	34660.14	PENDING	N/A	TRUE
BKG-003350	hunter weber	erinsmith@smith-kim.net	1081	San Fran	30/04/2024	12/06/2024	43	94.88	4079.84	paid	N/A	FALSE
BKG-015982	Elijah Osborne	madisongray@jacobson.org	1021	San Francisco	16/04/2025	15/05/2025	29	464.95	13483.55	Paid		y
BKG-019275	James Lester	hannah00@yahoo.com	1044	LA	20/10/2023	23/10/2023	3	466.35	1399.05	paid	4	No
BKG-018089			1074	SF	26/08/2023	08/03/2025	560	189.67	106215.2	Paid	6	FALSE
BKG-018753	robert reid	dcastro@hotmail.com	1058	nyc	23/12/2023	23/02/2025	428	268.87	115076.36	paid	2	TRUE
BKG-011503		jonesshawn@yahoo.com	1035	NYC	04/02/2024	12/02/2024	8	342.45	2739.6	pending	4	n
BKG-004058		michael88@gibson-burch.net	1083	New York	07/09/2024	01/12/2024	85	418.3	35555.5	paid	N/A	y
BKG-004874	LISA RICHARD		1006	LA	13/11/2023	02/01/2024	50	286.15	14307.5	Paid	-1	TRUE
BKG-019936	Jared Jordan	ellenlloyd@perez.biz	1044	San Fran	18/07/2024	02/04/2025	258	466.93	120467.94	PENDING		y
BKG-014055	tammie hall	ellisstephanie@gutierrez.net	1092	nyc	19/08/2023	12/02/2023	105	273.47	28714.35	Paid	4	TRUE
BKG-016852	ERICA SMITH	rodriguezjames@hayes.com	1095	NYC	19/11/2023	25/03/2025	492	278.03	136790.76	N/A	-1	y
BKG-018250	linda perry	taylorwheeler@yahoo.com	1081	Los Angeles	31/08/2023	03/01/2024	125	67.05	8381.25	Paid	4	y
BKG-015813	Margaret McClain		1099	LA	06/11/2023	05/11/2024	365	461.04	168279.6	pending	N/A	0
BKG-016414	John Werner DVM	brayjoseph@gmail.com	1040	San Fran	07/08/2024	12/04/2025	248	329.64	81750.72	paid	6	0
BKG-019453	robert cunningham	becky06@hotmail.com	1085	los angeles	24/09/2024	21/01/2025	119	457.42	54432.98	Paid	-1	Yes
BKG-001489	chad moore	douglas18@harmon.com	1067	los angeles	08/11/2024	08/11/2024	0	95.22	0		6	Yes

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BKG-012608	Mrs. Amanda Martinez	smartinez@olivia.com	1035	Los Angeles	21/03/2025	28/03/2025	7	189.12	1323.84	Unpaid	-1	TRUE
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BKG-014055	tammie hall	ellisstephanie@gutierrez.net	1095	NYC	19/11/2023	25/03/2025	492	278.03	136790.76	Paid	4	y
BKG-016852	ERICA SMITH	rodriguezjames@hayes.com	1081	Los Angeles	31/08/2023	03/01/2024	125	67.05	8381.25	pending	N/A	0
BKG-018250	linda perry	taylorwheeler@yahoo.com	1099	LA	06/11/2023	05/11/2024	365	461.04	168279.6	paid	-1	Yes
BKG-015813	Margaret McClain		1040	San Fran	07/08/2024	12/04/2025	248	329.64	81750.72			Yes
BKG-016414	John Werner DVM	brayjoseph@gmail.com	1085	los angeles	24/09/2024	21/01/2025	119	457.42	54432.98			TRUE
BKG-019453	robert cunningham	becky06@hotmail.com	1067	los angeles	08/11/2024	08/11/2024	0	95.22	0	PENDING	5	TRUE
BKG-001489	chad moore	douglas18@harmon.com	1090	nyc	28/02/2024	01/05/2025	428	480.46	205636.88	N/A		Yes
BKG-011967	John Torres	nmoss@gmail.com	1072	San Francisco	03/03/2025	03/03/2025	0	237.31	0		-1	0
BKG-009561	michael davis	johnsonsabrina@yahoo.com	1093	NYC	31/01/2024	27/09/2024	24	443.81	100773.6	Paid	6	1
BKG-001398	Bryce Gutierrez	vickiwood@gmail.com	1083	SF	08/03/2025	05-17-2025	24	443.81	22861.3	PENDING	5	n
BKG-013829	William Fisher	christopherbrown@yahoo.com	1090	NYC	14/08/2024	13/04/2025	31	443.81	36072.52	paid	6	TRUE
BKG-012431	Jennifer Ortiz	palmerrebecca@sanford-mckenzie.com	1067	los angeles	18/04/2024	27/04/2025	37	351.59	32852.16	N/A	1	FALSE
BKG-016160	MEGAN GRANT	sharon41@yahoo.com	1001	SF	17/03/2024	17/03/2024	0	351.59	0			FALSE
BKG-008688		debbieowens@willis.com	1059	SF	17/03/2024	17/03/2024	0	351.59	0	PENDING		No
BKG-001056	sheila miller	stacy94@robertson.info	1082	NYC	24/03/2024	30/04/2024	37	443.81	16420.97			n
BKG-014736	Sean Ross	james32@harris.biz	1081	New York	23/10/2024	07/11/2024	15	free	15000.00			1
BKG-000945	william howard	rachel86@brown.org	1075	San Fran	10/01/2024	31/12/2024	356	445.75	158697.00			TRUE
BKG-011832	Jasmine Savage	meganpalmer@joseph.com	1088	los angeles	21/12/2024	31/03/2025	100	430.22	43022.00		-1	FALSE
BKG-002826	DUSTIN POTTER	robert94@sloan-mcguire.com		SF	14/06/2024	02/12/2024	171	99.66	17041.86	N/A		

Note: This worksheet remains unprotected to allow for full transparency of the cleaning formulas and logic applied. To maintain data integrity, please avoid editing cells directly. For a finalized, analysis-ready version, please refer to the 'Cleaned Data' sheet.								
guest_name	name_extensions_check	cleaned_guest_name	email	cleaned_email	listing_id	countblanks_listing_id	listing_city	unique
CAROLYN SHAH Jermaine Huang amy rios	0	Unknown	nhoffman@scott.com	nhoffman@scott.com	1083	0	Los Angeles	Los Angel
	0	Carolyn Shah	achan@olson-nelson.com	achan@olson-nelson.com	1079		los angeles	San Fran
	0	Jermaine Huang	thomasadam@yahoo.com	thomasadam@yahoo.com	1036		San Fran	LA
	0	Amy Rios	reginald31@robertson-smith.com	reginald31@robertson-smith.com	1025		LA	new york
	0	Unknown		NA	1089		LA	San Franc
	0	Unknown	nichole43@gmail.com	nichole43@gmail.com	1038		LA	SF
Mrs. Amanda Martinez	0	Unknown	brianbrown@lynn-taylor.com	brianbrown@lynn-taylor.com	1066	0	San Fran	nyc
	1	Amanda Martinez	susanpham@livingston.com	susanpham@livingston.com	1035		Los Angeles	
shelby mason	0	Unknown	eharrison@mullins.com	eharrison@mullins.com	1038		new york	
	0	Shelby Mason	grahamamanda@sparks.com	grahamamanda@sparks.com	1089		San Fran	
hunter weber	0	Hunter Weber	erinsmith@smith-kim.net	erinsmith@smith-kim.net	1081		San Fran	
Elijah Osborne	0	Elijah Osborne	madisongray@jacobson.org	madisongray@jacobson.org	1021		San Francisco	
James Lester	0	James Lester	hannah00@yahoo.com	hannah00@yahoo.com	1044		LA	
	0	Unknown		NA	1074		SF	
robert reid	0	Robert Reid	dcastro@hotmail.com	dcastro@hotmail.com	1058		nyc	
	0	Unknown	jonesshawn@yahoo.com	jonesshawn@yahoo.com	1035		NYC	
LISA RICHARD	0	Unknown	michael88@gibson-burch.net	michael88@gibson-burch.net	1083		New York	
	0	Lisa Richard		NA	1006		LA	
Jared Jordan	0	Jared Jordan	ellenlloyd@perez.biz	ellenlloyd@perez.biz	1044		San Fran	
tammie hall	0	Tammie Hall	ellisstephanie@gutierrez.net	ellisstephanie@gutierrez.net	1092		nyc	
ERICA SMITH	0	Erica Smith	rodriguezjames@hayes.com	rodriguezjames@hayes.com	1095		NYC	
linda perry	0	Linda Perry	taylorwheeler@yahoo.com	taylorwheeler@yahoo.com	1081		Los Angeles	
Margaret McClain	0	Margaret McClain		NA	1099		LA	
John Werner DVM	0	John Werner Dvm	brayjoseph@gmail.com	brayjoseph@gmail.com	1040		San Fran	
robert cunningham	0	Robert Cunningham	becky06@hotmail.com	becky06@hotmail.com	1085		los angeles	
chad moore	0	Chad Moore	douglas18@harmon.com	douglas18@harmon.com	1067		los angeles	
John Torres	0	John Torres	nmoss@gmail.com	nmoss@gmail.com	1090		nyc	
michael davis	0	Michael Davis	johnsonsabrina@yahoo.com	johnsonsabrina@yahoo.com	1072		San Francisco	
Bryce Gutierrez	0	Bryce Gutierrez	vickiwood@gmail.com	vickiwood@gmail.com	1093		NYC	
William Fisher	0	William Fisher	christopherbrown@yahoo.com	christopherbrown@yahoo.com	1083		SF	
Jennifer Ortiz	0	Jennifer Ortiz	palmerrebecca@sanford-mckenzie.com	palmerrebecca@sanford-mckenzie.com	1090		NYC	
MEGAN GRANT	0	Megan Grant	sharon41@yahoo.com	sharon41@yahoo.com	1067		los angeles	
	0	Unknown	debbieowens@willis.com	debbieowens@willis.com	1001		SF	
sheila miller	0	Sheila Miller	stacy94@robertson.info	stacy94@robertson.info	1059		NYC	



Cleaned Data

Data Integrity Audit & Optimization: Standardizing 22,000 Multi-Source Hotel Records

Dataset: Dirty Data For Cleaning

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Author: Krithika

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BKG-013035	Unknown	nhoffman@scott.com	1083	Los Angeles	24/05/2023	12/12/2023	202	\$245.61	\$49,613.22	Pending	6	FALSE
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BKG-007591	Amy Rios	reginald31@robertson-smith.com	1025	Los Angeles	29/02/2024	16/03/2025	381	\$268.36	\$102,245.16	Pending	6	FALSE
BKG-000221	Unknown	NA	1089	Los Angeles	26/01/2025	22/02/2025	27	\$134.71	\$3,637.17	Pending		TRUE
BKG-017295	Unknown	nichole43@gmail.com	1038	Los Angeles	03/02/2025	16/02/2025	13	\$383.22	\$4,981.86	Unpaid		FALSE
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